

Evidence-Based Estimates of Outcome in Patients Irradiated for Intraocular Melanoma

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Background: Melanoma of the eye is the only potentially fatal ocular malignancy in adults. Until radiation therapy gained wide acceptance in the 1980s, enucleation was the standard treatment for the tumor. Long-term results after proton beam irradiation are now available.

Methods: We developed risk score equations to estimate probabilities of the 4 principal treatment outcomes—local tumor recurrence, death from metastasis, retention of the treated eye, and vision loss—based on an analysis of 2069 patients treated with proton beam radiation for intraocular melanoma between July 10, 1975, and December 31, 1997. Median follow-up in surviving patients was 9.4 years.

Results: Tumor regrowth occurred in 60 patients, and 95% of tumors (95% confidence interval, 93%-96%) were

controlled locally at 15 years. Risk scores were developed for the other 3 outcomes studied. Overall, the treated eye was retained by 84% of patients (95% confidence interval, 80%-87%) at 15 years. The probabilities for vision loss (visual acuity worse than 20/200) ranged from 100% to 20% at 10 years and for death from tumor metastases from 95% to 35% at 15 years, depending on the risk group.

Conclusions: High-dose radiation treatment was highly effective in achieving local control of intraocular melanomas. In most cases, the eye was salvaged, and functional vision was retained in many patients. The mortality rate was high in an identifiable subset of patients who may benefit from adjuvant therapies directed at microscopic liver metastases.

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UVEAL MELANOMA is an uncommon intraocular malignancy, occurring in fewer than 2000 new patients in the United States each year.¹ It is the only intraocular disease that is commonly fatal. Until the introduction of radiation therapy in the 1960s, enucleation was the standard therapy for ocular melanoma. However, a variety of alternative approaches have been attempted in an effort to salvage the eye while eradicating the primary tumor.²⁻¹⁰ Among these, radiation therapy has the widest applicability. The 2 major forms of radiation therapy used in the treatment of these tumors are brachytherapy, in which radiation is delivered through a radioactive plaque sutured to the sclera over the base of the melanoma, and external beam irradiation using proton and helium ion beams.

Both modalities deliver a large dose to the portion of the eye involved by tumor but can limit the dose delivered to the rest of the eye and to adjacent normal structures. In this article, we report treatment outcomes in patients treated with protons,

charged particles whose physical properties offer several advantages over other forms of external beam irradiation. Protons deposit most of their energy near their end of range, the so-called Bragg peak effect, resulting in a uniform and highly localized dose being deposited largely in the tumor. The beam energy can be varied to limit the dose that reaches normal tissues distal to the tumor. The beam can also be collimated to match the tumor profile, with a sharp lateral dose falloff of approximately 1.1 mm (80% to 20%).^{11,12} These characteristics produce a dose distribution that conforms to the tumor, sparing adjacent healthy tissue. These advantages are especially critical for the treatment of intraocular melanomas, which have a predilection to develop near the macula and optic disc.¹³ The near-uniform dose delivered to the tumor, and the sharp reduction of dose outside the treated area, should increase the therapeutic ratio of tumor control to normal tissue complications in these patients compared with other radiation modalities.

Results in patients with intraocular melanoma treated over 2 decades at the Har-

Table 1. Cumulative Baseline Rates of End Point–Free Survival (S_t in Equation 2)*

Year	Metastatic Death Rate	Enucleation Rate	Vision Loss Rate
1	0.99997	0.99903	0.98927
2	0.99989	0.99781	0.97747
3	0.99975	0.99676	0.96493
4	0.99964	0.99579	0.95206
5	0.99951	0.99510	0.93959
6	0.99938	0.99455	0.93018
7	0.99929	0.99340	0.92623
8	0.99923	0.99358	0.91632
9	0.99920	0.99333	0.91179
10	0.99913	0.99308	0.90505
11	0.99907	0.99274	0.90368
12	0.99904	0.99205	0.90207
13	0.99899	0.99136	0.90207
14	0.99897	0.98961	0.89964
15	0.99894	0.98961	0.88325

*These rates can be used in conjunction with Equation 2 to calculate the likelihood of a given end point at the times indicated for a given risk score. Further information on the risk score formulations in this article is available from the authors.

vard Cyclotron Laboratory in Cambridge, Mass, indicate that proton beam irradiation is highly effective in achieving local control in these tumors and that most patients retain the treated eye. However, functional results and survival times are highly variable, depending on patient attributes and tumor characteristics. We developed risk score equations for predicting the probability of these outcomes. The ability to predict these events will potentially benefit both patient and physician in defining treatment strategies and providing rationales for new clinical trials designed to optimize patient outcome.

METHODS

We analyzed 2069 consecutive patients from the United States and Canada, treated by a single ophthalmologist (E.G.), with unilateral choroidal and/or ciliary body melanomas, without evidence of systemic metastasis before treatment, and treated with proton radiation at the Harvard Cyclotron Laboratory between July 10, 1975, and December 31, 1997.

Details of proton treatment of intraocular melanomas have been described in previous articles.^{9,12,14,15} The radiation dose was 70 cobalt gray equivalents (CGE) delivered in 5 equal fractions over 7 to 10 days in most patients (94%); the remaining patients were treated with slightly higher or lower doses ranging from 50 to 100 CGE. (A CGE is the physical dose in gray multiplied by a relative biological effectiveness factor of 1.10. This is judged to be the dose that would be biologically equivalent to the numerically same dose if delivered by a cobalt 60 photon beam.) All patients were fully informed of the risks and benefits of proton therapy for treatment of the eye tumor. Mortality surveillance was completed through June 30, 1999, and ocular outcomes were ascertained through April 30, 1998. Patients were followed on a regular basis at the Massachusetts Eye and Ear Infirmary or by their referring ophthalmologists. Vital status was also updated from the National Death Index and the Social Security Death Index.¹⁶⁻¹⁸ A total of 735 deaths were documented; 408 were due to metastatic melanoma. Median follow-up among 1334 surviving patients was 9.4 years (range, 10 months to 24 years). Cause of death was established by medical documentation in 76% of patients, from death certificates

in 17%, or from next of kin in 7%. Median ophthalmic follow-up after treatment was 4.6 years (range, 3 months to 20 years) and was available in 1945 patients (94%).

STATISTICAL ANALYSIS

Rates of occurrence and prognostic factors were determined for 4 outcomes: (1) local regrowth of the tumor, (2) tumor-related mortality, (3) enucleation, and (4) vision loss (to visual acuity worse than 20/200). For all time-related analyses, patients were followed from the end of irradiation until either the date of the event or, in censored observations, the date of the last dilated examination for tumor recurrence, the last visual acuity measurement for visual outcome before April 30, 1998, and the earlier of the date of the last previous contact or June 30, 1999, for tumor-related mortality.

We used multivariate Cox proportional hazards regression models¹⁹ to identify statistically significant prognostic factors independently related to each outcome except for local recurrence, which was too rare for multivariate analysis. In addition to contributions from individual factors, cross-products were examined for possible interactions; none of significance were found. The proportional hazards assumption was examined using the graphical methods of Hess²⁰ and Garrett²¹ and tests of the proportional hazard assumption based on Schoenfeld residuals²²; no violations were noted.

RISK SCORE ANALYSIS

To estimate predicted probabilities of each event at particular points after treatment, we developed risk score formulations based on the method of Cochran and associates.²³ The baseline survival rates and coefficients (log hazard ratios) of each prognostic factor were estimated by the most parsimonious Cox proportional hazards model that included all identified significant risk factors. The risk score for a given individual is estimated by summing, over all significant predictive factors, the product of the level x_i of the i^{th} risk factor multiplied by its corresponding coefficient β_i . That is, for each outcome considered:

(1)

$$\text{Risk Score} = \sum_i x_i \beta_i$$

To obtain the SE for the risk score, we regressed estimated risk score values on outcome in a univariate Cox model; a robust method was used to calculate the variance-covariance matrix.²⁴

Given the risk score, the probability of occurrence of each end point as a function of time since treatment can be deduced from the baseline survival rate, which is an output parameter of the Cox analysis. That probability, $P_{\text{end point}}$, is given by the following formula:

(2)

$$P_{\text{end point}} = 1 - S_t^{\exp(\text{risk score})}$$

where S_t is the baseline survival rate for the end point of interest. Cumulative baseline rates of end point–free survival (S_t in Equation 2) for each year after treatment are given in **Table 1**. Given these baseline survival rates, the 95% confidence interval (CI) for $P_{\text{end point}}$ can be calculated as follows:

(3)

$$\text{Lower Bound for } P_{\text{end point}} = 1 - S_t^{\exp[\text{risk score} (1 - 2 \times \text{SE})]}$$

(4)

$$\text{Upper Bound for } P_{\text{end point}} = 1 - S_t^{\exp[\text{risk score} (1 + 2 \times \text{SE})]}$$

For each outcome, we estimated actuarial rates at 5, 10, and 15 years after treatment according to risk score category and 95% CIs. Patients with evidence of extrascleral extension at presentation (4% of the series) were excluded from analyses

of tumor death, as these patients were at unusually high risk of dying of metastases; the 10-year rate for death from tumor-related causes in these patients was 65% (95% CI, 52%-78%) compared with 23% (95% CI, 21%-25%) for all other patients.

RESULTS

PATIENT AND TUMOR CHARACTERISTICS

An equal number of men and women were treated (49% and 51%, respectively), and the average age at treatment was 61 years (range, 11-93 years). Most patients (72%) had symptoms such as recent loss in vision or photopsia; the remaining patients were diagnosed during routine eye examination. Detection of the tumor on routine examination was more common in older patients ($P<.001$) and women ($P<.001$).

There was no predilection for the tumor to occur in either eye (51% in the right eye and 49% in the left eye). In 68% of patients, the tumor had a posterior margin within 2 disc diameters (approximately 3.0 mm) of the optic disc or macula. The median basal diameter of the tumor was 13.2 mm, and the median tumor height was 5.3 mm. Median visual acuity at presentation was 20/40; 14% of patients had substantial vision loss (visual acuity worse than 20/100), generally related to large retinal detachment or tumor encroaching on the macula or optic nerve, whereas 20% had visual acuity of 20/20 or better.

OUTCOME-SPECIFIC FINDINGS

Local Control

Local recurrence of the tumor was documented by sequential ultrasound or fundus photographs in 45 patients; in 15 additional patients, the eye was promptly enucleated because of suggestion of tumor regrowth. Among the 45 documented cases, 22 represented marginal recurrence, 9 showed vertical growth in the main mass of the tumor, and the remaining 14 represented extrascleral extensions ($n=8$) or outgrowth of possible "ring" melanomas ($n=6$).

Death From Tumor-Related Causes

A total of 408 patients (20%) died of metastatic melanoma, and 17 more were alive with metastasis by July 1999. Patients died of metastasis 3 months to 16 years after proton beam irradiation. Annual death rates from metastases peaked 3 to 6 years after treatment at approximately 4 deaths per 100 patients per year.

Retention of the Treated Eye

The eyes of 179 patients (9%) were enucleated, primarily owing to neovascular glaucoma (46%), blind uncomfortable eye (31%), or local recurrence of the tumor (23%).

Visual Loss

Approximately 14% of patients had significant loss of vision (visual acuity worse than 20/100); the vision of these

Table 2. Univariate Analysis of Prognostic Factors for Local Failure*

Risk Factor	RR (95% CI)	P Value
LTD: 1 mm	1.19 (1.06-1.32)	.002
Anterior margin		
Posterior to equator	Referent	...
Anterior to equator	0.58 (0.12-2.71)	.48
Ciliary body and iris	3.38 (1.38-8.26)	.008
Tumor height: 1 mm	1.13 (0.98-1.29)	.08
Iris color: brown vs green-blue	0.19 (0.02-1.39)	.10
Symptom: present vs absent	2.59 (0.77-8.74)	.13
Extrascleral extension: present vs absent	1.33 (0.86-2.06)	.20

*RR indicates relative risk; CI, confidence interval; and LTD, largest tumor diameter. Statistically nonsignificant factors ($P>.20$) are not listed.

Table 3. Risk Factors for Disease-Related Mortality for Patients Without Evidence of Extrascleral Extension at Baseline Examination and an Example of Risk Score Estimation*

Prognostic Factor	Coefficient	P Value	Sample Risk Score Calculation	
			Value	Contribution to Risk Score
Tumor diameter: 1 mm	0.191	<.001	15	2.87
Age at treatment: 1 y	0.025	<.001	60	1.50
Tumor pigment: minimal (0) vs moderate and heavy (1)	0.873	<.001	1	0.87
Symptoms: none (0) vs any (1)	0.575	<.001	1	0.58
Tumor origin: choroidal (0) vs ciliary body (1)	0.512	.004	0	0
Iris color: brown (0) vs green-blue (1)	0.294	.03	1	0.29
Risk score				6.11

*Robust SE of the coefficient of the estimated risk score is 0.06185.

patients rarely improved, and they generally lost additional lines of visual acuity after radiation treatment. Most of these patients had only hand motion or worse visual acuity 10 years after irradiation. The prognosis was better in patients with good vision at the time of presentation; therefore, we limited assessment of risk factors to patients with baseline visual acuity of at least 20/100.

ANALYSIS OF PROGNOSTIC FACTORS

Table 2 presents the results of a univariate analysis of factors associated with local failure (combining documented and suspected cases). **Tables 3, 4, and 5** present the results of multivariate analyses of prognostic factors for the other end points considered (death from tumor-related causes, loss of the treated eye, and loss of vision to visual acuity worse than 20/200). Table 3, in addition, provides an example of the computation of the risk score for a "prototypical" patient for the end point of death from tumor-related causes. In this patient, the cumulative 3-year tumor-specific death rate given the risk

Table 4. Risk Factors for Eye Loss*

Prognostic Factor	Coefficient	P Value
Tumor height: 1 mm	0.211	<.001
Distances from the macula and optic disc: >2 DD either (0) or ≤2 DD both (1)	0.703	<.001
Tumor diameter: 1 mm	0.068	.02
Tumor pigment: minimal and moderate (0) or heavy (1)	0.377	.04
Tumor shape: dome and mushroom (0) or bilobed and diffuse (1)	0.667	.046

*Robust SE for the coefficient of the estimated risk score is 0.07420.
DD indicates disc diameter.

Table 5. Risk Factors for Vision Loss to Visual Acuity Worse Than 20/200 for Patients With Initial Visual Acuity of 20/100 or Better*

Prognostic Factor	Coefficient	P Value
Distances from the macula and optic disc >2 DD from both	0	...
≤2 DD from either	0.668	<.001
≤2 DD from both	1.391	<.001
Tumor height: 1 mm	0.212	<.001
Visual acuity at baseline 20/25 or better	0	...
20/30-20/40	0.365	<.001
20/50 or worse	0.613	<.001
Retina detachment: <1 quadrant (0) or ≥1 quadrant (1)	0.326	<.001
Diabetes mellitus history: no (0) or yes (1)	0.390	.004
Tumor diameter: 1 mm	0.030	.04

*Robust SE of the coefficient of the estimated risk score is 0.04590.
DD indicates disc diameter.

score of 6.11 is 10.6% (95% CI, 5.1%-21.3%). These numbers are derived as follows: if $S_3=0.99975$ (from Table 1), then the probability of death by 3 years after radiation = $1 - 0.99975^{\exp(6.11)} = 10.6\%$. The robust SE for the coefficient of the estimated risk score is 0.06185; the 95% CI for the death rate by 3 years after treatment is calculated as follows:

$$\text{Lower Bound} = 1 - 0.99975^{\exp[6.11 \times (1 - 2 \times 0.06185)]} = 5.1\%$$

$$\text{Upper Bound} = 1 - 0.99975^{\exp[6.11 \times (1 + 2 \times 0.06185)]} = 21.3\%$$

In **Table 6**, estimated probabilities for each outcome after 5, 10, and 15 years after irradiation are shown according to risk score categories.

ACTUARIAL LIKELIHOOD OF EXPERIENCING EACH END POINT

Figures 1, 2, 3, and 4 show the results of Kaplan-Meier analyses for the cumulative probability of a patient experiencing 1 of the 4 end points after proton beam irradiation. In Figures 2, 3, and 4, for the end points death from tumor-related causes, loss of the treated eye, and

Table 6. Estimated Probabilities and 95% Confidence Intervals (CIs) for Each Outcome 5, 10, and 15 Years After Irradiation, According to Risk Score

Risk Score	Rate (95% CI), %		
	5 y	10 y	15 y
Metastatic Death			
4.0	3 (2-4)	5 (3-7)	6 (3-9)
4.5	4 (2-7)	8 (4-13)	9 (5-15)
5.0	7 (4-13)	12 (7-21)	15 (8-25)
5.5	11 (6-21)	19 (10-34)	23 (12-40)
6.0	18 (9-34)	30 (15-52)	35 (18-59)
6.5	28 (14-52)	44 (23-73)	51 (27-79)
7.0	42 (20-72)	61 (33-90)	69 (39-94)
7.5	59 (30-89)	79 (46-98)	85 (53-99)
8.0	77 (42-98)	93 (62-100)	96 (69-100)
8.5	91 (57-100)	99 (78-100)	99 (84-100)
9.0	98 (73-100)	100 (90-100)	100 (94-100)
9.5	100 (87-100)	...	...
Eye Loss			
2.0	4 (3-5)	5 (4-7)	7 (6-10)
2.5	6 (4-8)	8 (6-12)	12 (8-17)
3.0	9 (6-14)	13 (9-20)	19 (13-28)
3.5	15 (9-24)	21 (13-32)	29 (19-44)
4.0	24 (14-38)	32 (19-50)	43 (27-64)
4.5	36 (20-58)	46 (27-70)	61 (38-84)
5.0	52 (29-78)	64 (39-89)	79 (52-96)
5.5	70 (41-93)	82 (53-98)	92 (68-100)
6.0	86 (56-99)	94 (68-100)	99 (82-100)
6.5	96 (71-100)	99 (83-100)	100 (93-100)
7.0	100 (85-100)	100 (93-100)	..
Vision Loss			
0.5	10 (9-10)	15 (15-16)	19 (18-19)
1.0	16 (14-17)	24 (22-26)	29 (26-31)
1.5	24 (22-27)	36 (32-40)	43 (38-47)
2.0	37 (32-42)	52 (46-59)	60 (53-67)
2.5	53 (45-62)	70 (62-78)	78 (70-85)
3.0	71 (61-81)	87 (78-93)	92 (85-96)
3.5	87 (78-94)	96 (91-99)	98 (95-100)
4.0	97 (91-99)	100 (98-100)	100 (99-100)
4.5	100 (98-100)	...	...

loss of vision to visual acuity worse than 20/200, respectively, actuarial plots are shown by increasing risk score.

COMMENT

We present an evidence-based risk score approach for estimating probabilities of critical outcomes in patients undergoing conservation treatment for intraocular melanoma. Whereas conventional methods based on Kaplan-Meier curves are at the group level, this method allows clinicians to assess prognosis after treatment at the individual patient level. The Cox regression model has been widely used in assessment of relative risk in ophthalmologic research, but it has not been used previously to estimate the probability of outcomes. For clinicians, the absolute risk (probability) of an adverse event may be more useful information than the relative risk. Such outcome estimates can be useful to patients and clinicians when weighing treatment options. In addition, the technique can identify patients at higher risk of dying of metastasis who may be optimal candidates for adjunctive therapy^{23,25-27} or those

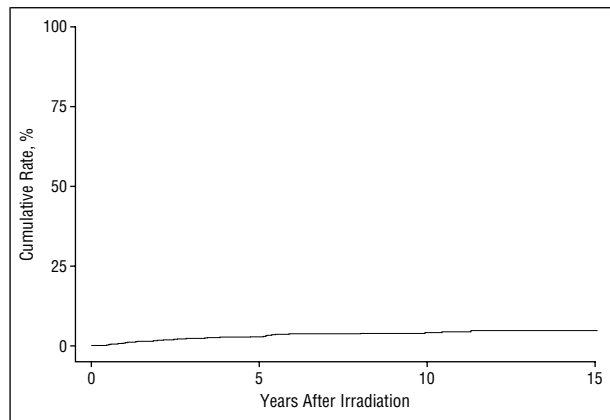


Figure 1. Results of Kaplan-Meier analyses for the cumulative probability of local recurrence of the tumor after proton irradiation. The numbers of patients followed (and the numbers developing recurrence) were 2062 (49) for 0 to 4.9 years, 979 (9) for 5.0 to 9.9 years, 377 (2) for 10.0 to 14.9 years, and 47 (2) for 15.0 years or more after irradiation.

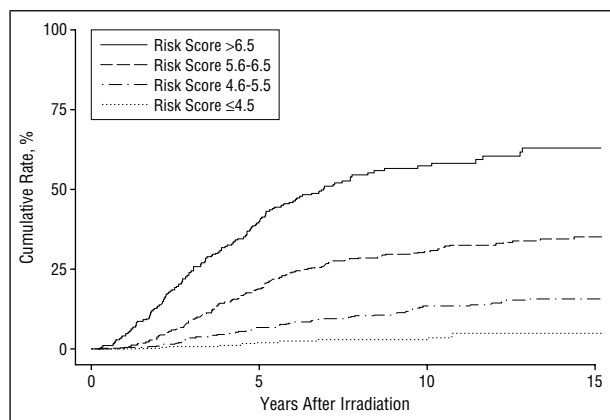


Figure 2. Results of Kaplan-Meier analyses for the cumulative probability of death from tumor-related causes. Patients with extraocular extension are excluded. The numbers of patients followed (and the numbers dying) were 2069 (265) for 0 to 4.9 years, 1366 (116) for 5.0 to 9.9 years, 730 (25) for 10.0 to 14.9 years, and 222 (25) for 15.0 years or more after irradiation.

experiencing higher rates of eye or vision loss for dose-searching, fractionation, or other studies that might reduce treatment-related morbidity. Data used to derive these estimates result from the largest and most comprehensive analysis to date of the functional outcome and mortality experience of patients undergoing conservation therapy for intraocular melanoma. Results indicate that most patients experience a good outcome with respect to local control and eye retention, and many retain useful eye function for long periods after treatment. However, there is considerable heterogeneity among patients in tumor size and other factors that predict patient outcome.

Local control rates exceeding 95% at 10 years are among the highest yet reported for any solid tumor. Causes of documented local failures in 45 patients included marginal failures, those due to outgrowth of possible ring melanomas, and true in-field recurrences. The first category may represent failure to accurately model tumor extent in treatment planning, improper treatment delivery, or patient inability to adequately fixate the eye for treatment. Only a few tumors (9 confirmed) were resistant to the prescribed radiation dose. These were larger tumors, which

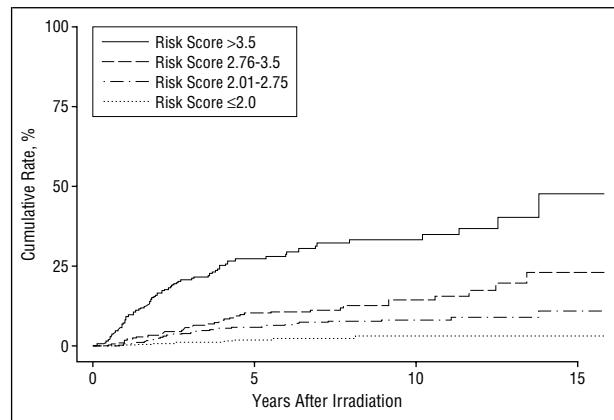


Figure 3. Results of Kaplan-Meier analyses for the cumulative probability of enucleation of the eyes. The numbers of patients followed (and the numbers undergoing enucleation) were 2058 (143) for 0 to 4.9 years, 979 (23) for 5.0 to 9.9 years, 383 (10) for 10.0 to 14.9 years, and 48 (0) for 15.0 years or more after irradiation.

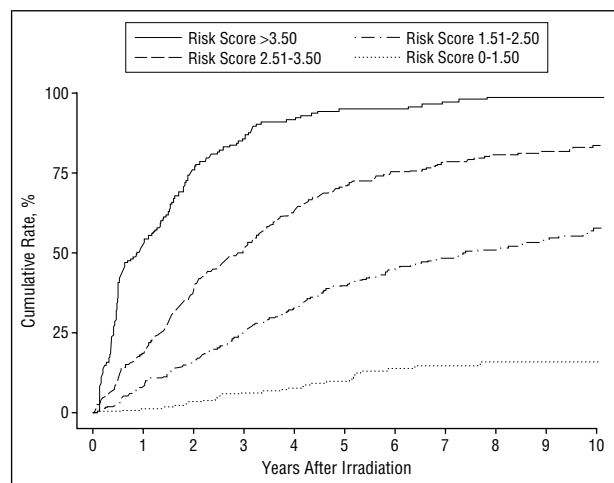


Figure 4. Results of Kaplan-Meier analyses for the cumulative probability of vision loss to visual acuity worse than 20/200 among patients with good starting vision (visual acuity 20/100 or better). The numbers of patients followed (and the numbers losing vision) were 1462 (655) for 0 to 4.9 years and 416 (82) for 5.0 to 9.9 years after irradiation. In later follow-up, these numbers were 127 (7) for 10.0 to 14.9 years and 22 (2) for 15.0 years or more after irradiation (not shown).

may have outgrown the local blood supply and become hypoxic, rendering them less sensitive to irradiation. However, most tumors in this size range were controlled, so the doses used were unlikely to be deficient.

The exceptionally high control rates at 70 CGE led us to undertake a randomized dose-searching clinical trial to evaluate the safety and efficacy of reducing the dose to 50 CGE with the same fractionation and overall treatment interval. The early results of the trial²⁸ suggest that the lower dose is sufficient for local control. However, the functional results were not improved. We are now planning a trial to determine whether hyperfractionation might reduce radiation damage to visual structures and improve functional results while achieving comparable levels of local tumor control.

Irradiation-related complications involving the anterior segment that contribute to a decline in visual acuity or eye loss include cataract^{29,30} and neovascular glau-

coma.³¹⁻³³ Cataract extraction is feasible in these patients and can restore vision in some.³⁴⁻³⁶ We found that pan-retinal photocoagulation can substantially reduce the severity of neovascular glaucoma and reduce the risk of enucleation by half (E.G., A.M.L., W.L., K.M.E., unpublished data, 2000). Posterior segment complications include radiation maculopathy³⁷⁻³⁹ and papillopathy.^{39,40} Radiation maculopathy affects central vision, whereas radiation papillopathy can lead to total loss of vision.

The evidence-based risk score formulations presented in this article should provide accurate prediction of these outcomes and should allow development of clinical trials aimed at reducing irradiation-related morbidity. The large size of the study population, the comprehensive follow-up, and the large number of events provide robust results that should be applicable to most patients diagnosed as having these tumors. Figures 1 through 4 demonstrate an increasing incidence of outcomes corresponding with the risk score, providing validation for the risk score estimates derived within this population.

Results for brachytherapy of intraocular melanoma have been reported⁴¹⁻⁴³ and suggest that this radiation modality may also offer a good prognosis for vision in selected patients.^{42,43} However, local recurrence rates are often high,^{44,45} particularly for tumors in close proximity to the optic nerve, possibly because of technical difficulties in the placement of plaques posteriorly. Generally, brachytherapy is not suitable for highly elevated tumors (>10 mm) because of the excessive scleral dose needed to successfully irradiate the tumor apex. The major limiting factor associated with charged particle irradiation in the United States is that only 3 clinical centers offer the treatment: the Harvard Cyclotron Laboratory, the University of California at Davis, and Loma Linda Medical Center, Loma Linda, Calif. Outside of the United States, the largest experience with proton beam irradiation has been accumulated at the Paul Scherrer Institute in Villigen, Switzerland.¹⁰

Although preservation of the eye and useful vision in many patients are the obvious advantages of conservation treatment with radiation compared with enucleation, questions have been raised regarding its efficacy in limiting metastatic spread of the tumor. In the present series, rates of tumor-related mortality are comparable to those reported in the literature for enucleation.^{46,47} Observational studies⁴⁸⁻⁵⁰ comparing irradiation with enucleation have also indicated no significant differences in survival. In the Collaborative Ocular Melanoma Study,⁵¹ a multicenter randomized clinical trial sponsored by the National Eye Institute (Bethesda, Md), overall 5-year mortality rates were comparable—at approximately 20%—in patients treated by enucleation and by plaque radiotherapy (iodine 125).

In conclusion, evidence-based risk score estimates derived from results analysis in a large series of patients predict specific outcomes in individual patients or in defined groups of patients. This ability can aid in treatment decisions for individual patients and in designing clinical trials. The results show that proton irradiation is highly successful in achieving local control of intraocular melanomas. Vision-threatening complications, including radiation maculopathy and papillopathy, are com-

mon in these patients because the tumors arise in proximity to the macula and optic nerve. However, many patients maintain some degree of function in the eye for long periods after treatment. Overall rates of metastatic disease are comparable to those observed after enucleation; thus, enucleation should be limited to patients with large tumors in whom the eye is unlikely to be salvaged by irradiation.

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